

MatMul-Free Model for Image Classification

PROJECT DESCRIPTION

Neural networks rely heavily on multiplications, which are computationally more expensive than additions and subtractions. Matrix Multiplication Free (MMF) networks replace multiplications with additions and subtractions, offering the potential of reducing energy consumption.

The thesis investigates the feasibility of MMF networks applied to image classification. Two architecturally equivalent models are designed and trained under identical conditions on the CIFAR-10 dataset: a full-precision Matrix Multiplication Model and a MMF Model. Then, several experimental runs are conducted covering different optimization techniques to improve the MMF Model.

Final results show that the MMF network is trainable, achieving an accuracy of 67.1% in validation data. However, the full-precision model achieves 85.6% accuracy. Closing this performance gap remains an open challenge that motivates further architectural innovation.

Matrix Multiplication Layer (standard)

$$W \in \mathbb{R}^{d \times m} \quad x \in \mathbb{R}^d \quad y \in \mathbb{R}^m$$

$$y_i = xW = \sum_{j=1}^d x_j W_{ij} \quad \text{for } i = 1, 2, \dots, m$$

Matrix Multiplication Free Layer (MMF)

$$W \in \mathbb{R}^{d \times m} \quad x \in \mathbb{R}^d \quad \tilde{Y} \in \mathbb{R}^m$$

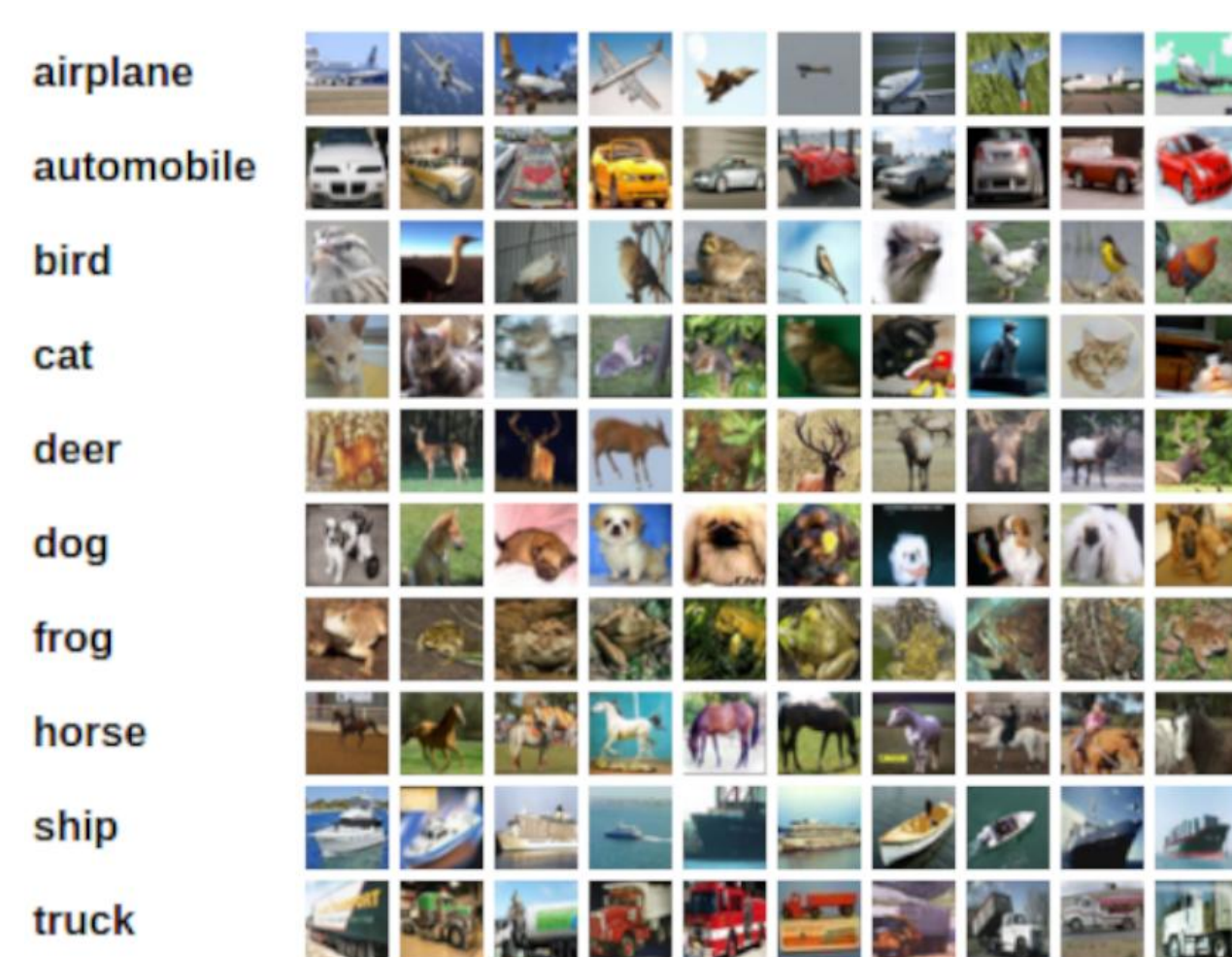
$$\tilde{Y}_i = \sum_{j=1}^d x_j \tilde{W}_{ij}$$

$$\tilde{W}_{ij} \in \{-1, 0, +1\}$$

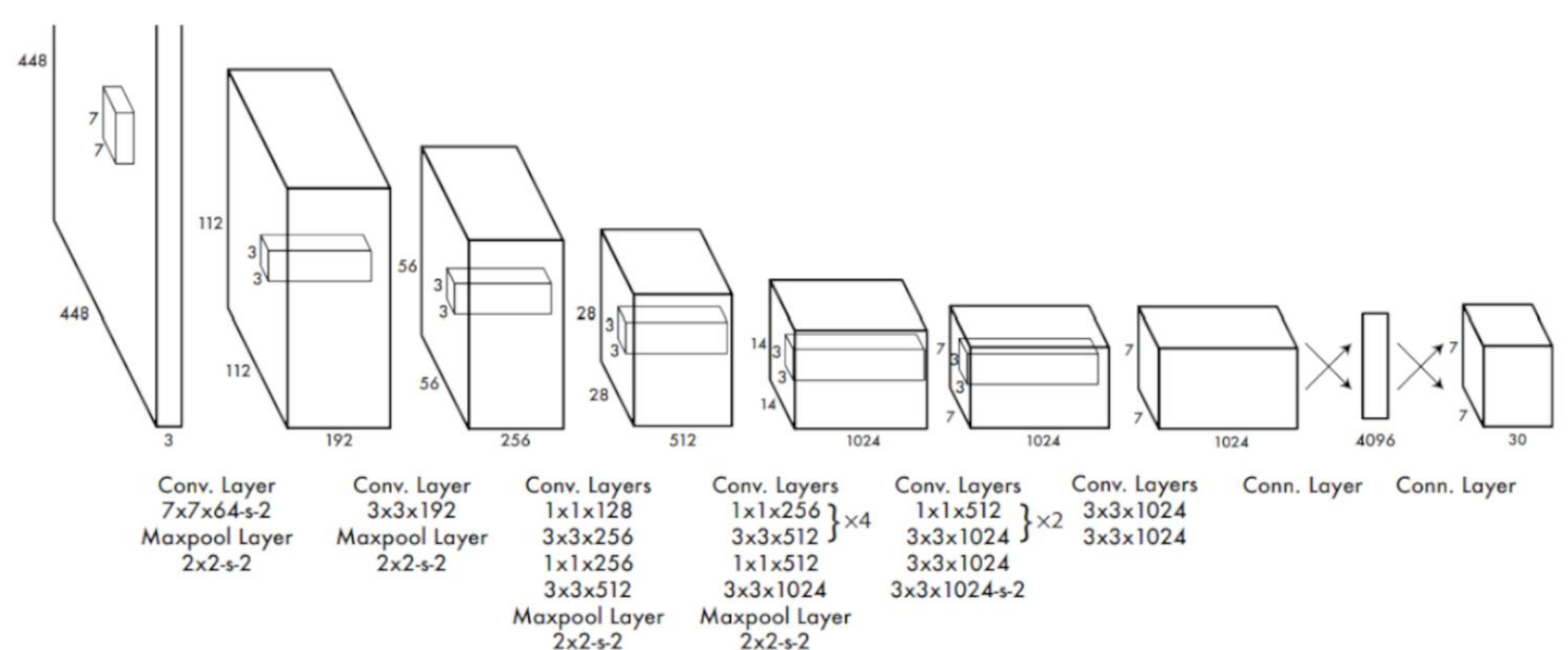
$$x_j \tilde{W}_{ij} = \begin{cases} x_j, & \text{if } \tilde{W}_{ij} = 1, \\ 0, & \text{if } \tilde{W}_{ij} = 0, \\ -x_j, & \text{if } \tilde{W}_{ij} = -1. \end{cases}$$

$$\tilde{Y}_i = \sum_{j: \tilde{W}_{ij}=1} x_j - \sum_{j: \tilde{W}_{ij}=-1} x_j$$

CIFAR-10 Classification Dataset



YOLOv1-based Model Architecture



Results: Validation Accuracy % for Baseline vs Best MMF Model

